

CYBER SECURITY INTERNSHIP REPORT

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Major: Computer Science Engineering (Cybersecurity)

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Domain: Ethical Hacking

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Projects and Tasks (Intermediate Level)

Assessment 3: Anonymous Communication Using Tor Network

1. Introduction

The main objective of this task is to understand how anonymous communication works using the Tor network and to implement IP rotation for maintaining operational security.

In cybersecurity, it is important to hide the real identity of the system while performing analysis or testing. In this task, I used Tor along with Proxymal and browser configuration to route traffic anonymously. I also verified anonymity using IP and DNS leak testing tools.

2. Understanding Tor Network

Tor (The Onion Router) is a privacy-focused network that hides user identity by routing traffic through multiple nodes.

How Tor Works:

- Data is encrypted in multiple layers
- It passes through:
 - Entry Node
 - Relay Node
 - Exit Node
- The website only sees the **Exit Node IP**, not the real IP

3. Lab Environment

- Operating System: Kali Linux
- Tools Used:
 - Tor Service
 - Proxychains
 - Firefox Browser
 - Curl
- Internet Connection: Required

4. Step-by-Step Implementation

4.1 Installing Tor

```
sudo apt update  
sudo apt install tor -y
```

4.2 Starting Tor Service

```
sudo service tor@default start
```

Tor was successfully started and initialized.

4.3 Verifying Tor Connection

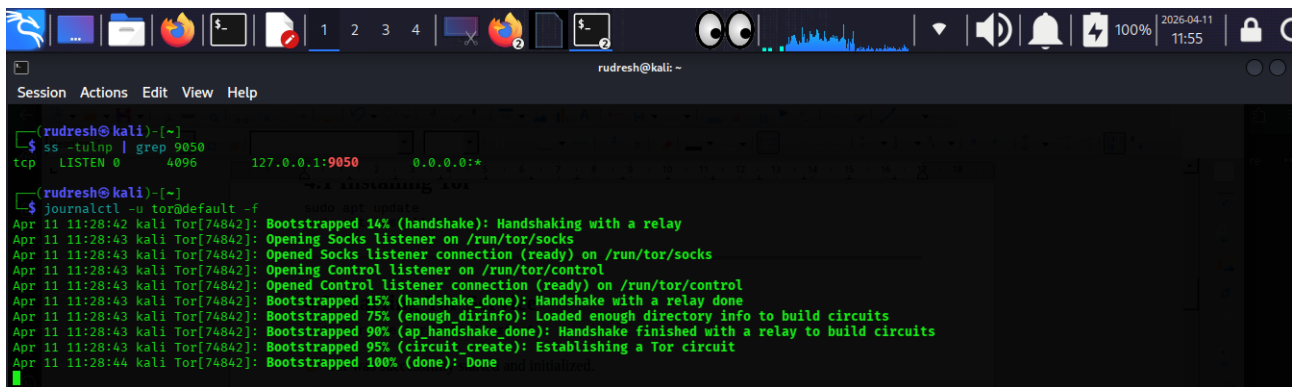
```
journalctl -u tor@default -f
```

Tor logs showed:

```
Bootstrapped 100% (done)
```

This confirms Tor network is fully connected.

Output:

A terminal window on a Kali Linux system showing the startup of the Tor service. The user runs 'journalctl -u tor@default -f' to follow the logs. The logs show the Tor daemon starting, opening Socks and Control listeners, and bootstrapping the network. The bootstrapping progress is shown as a series of steps: 14% (handshake), 15% (handshake done), 75% (enough dirinfo), 90% (ap handshake done), 95% (circuit create), and finally 100% (done). The terminal also shows the output of 'ss -tulnp | grep 9050' which indicates that the Socks listener is listening on 127.0.0.1:9050.

```
(rudresh@kali)~$ ss -tulnp | grep 9050
tcp        LISTEN    0      4096      127.0.0.1:9050      0.0.0.0:*

(rudresh@kali)~$ journalctl -u tor@default -f
Apr 11 11:28:42 kali Tor[74842]: Bootstrapped 14% (handshake): Handshaking with a relay
Apr 11 11:28:43 kali Tor[74842]: Opening Socks listener on /run/tor/socks
Apr 11 11:28:43 kali Tor[74842]: Opened Socks listener connection (ready) on /run/tor/socks
Apr 11 11:28:43 kali Tor[74842]: Opening Control listener on /run/tor/control
Apr 11 11:28:43 kali Tor[74842]: Opened Control listener connection (ready) on /run/tor/control
Apr 11 11:28:43 kali Tor[74842]: Bootstrapped 15% (handshake done): Handshake with a relay done
Apr 11 11:28:43 kali Tor[74842]: Bootstrapped 75% (enough dirinfo): Loaded enough directory info to build circuits
Apr 11 11:28:43 kali Tor[74842]: Bootstrapped 90% (ap handshake done): Handshake finished with a relay to build circuits
Apr 11 11:28:43 kali Tor[74842]: Bootstrapped 95% (circuit create): Establishing a Tor circuit
Apr 11 11:28:44 kali Tor[74842]: Bootstrapped 100% (done): Done and initialized
```

4.4 Checking Tor Port

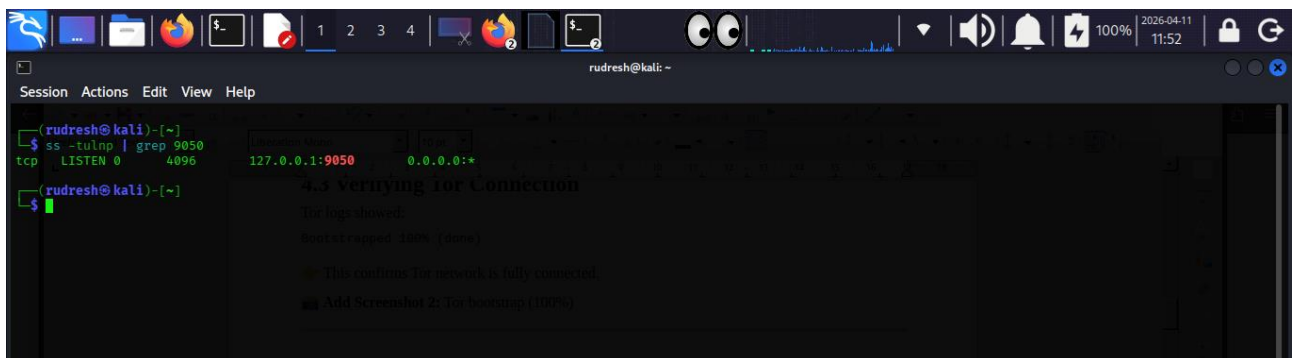
```
ss -tulnp | grep 9050
```

Output:

```
127.0.0.1:9050 LISTEN
```

This confirms SOCKS proxy is active.

Result:

A terminal window showing the Tor service status and a connection verification step. The user runs 'ss -tulnp | grep 9050' again, showing the Socks listener is listening on 127.0.0.1:9050. Below this, there is a section titled '4.3 verifying Tor Connection' which shows the logs from the previous screenshot, confirming that the Tor network is fully connected. The terminal also shows the output of 'ss -tulnp | grep 9050' which indicates that the Socks listener is listening on 127.0.0.1:9050.

```
(rudresh@kali)~$ ss -tulnp | grep 9050
tcp        LISTEN    0      4096      127.0.0.1:9050      0.0.0.0:*

(rudresh@kali)~$ journalctl -u tor@default -f
4.3 verifying Tor Connection
The logs showed:
Bootstrapped 100% (done)
This confirms Tor network is fully connected.
Add Screenshot 2: Tor bootstrap (100%)
```

4.5 Installing Proxychains

```
sudo apt install proxychains -y
```

4.6 Configuring Proxychains

Edited:

```
sudo nano /etc/proxychains.conf
```

Changes made:

- Enabled `dynamic_chain`
- Enabled `proxy_dns`

- ```
socks5 127.0.0.1 9050
```

```
proxychains4 curl http://httpbin.org/ip
```

```

(rudresh@kali)-[~]
$ proxychains4 curl http://httpbin.org/ip
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:80 ... OK
{
 "origin": "185.220.101.63"
}

(rudresh@kali)-[~]
$
185.220.100.130 dns124au31.netze.de F3 Netze e.V. Germany
185.220.101.31 berlin01app-ext.eu Stiftung Erneuerbare Freiheit Brandenburg, Germany
185.241.208.136 ns2.rdp.rs 1037 Services Warsaw, Poland
192.42.116.136 None Church of Cyberology Netherlands
192.42.116.38 None Church of Cyberology Netherlands
192.42.116.8 None Church of Cyberology Netherlands
192.42.116.88 None Church of Cyberology Netherlands
217.197.91.153 artikel10-in-berlin... Individual Network Berlin (IN-Berlin) e.V. Berlin, Germany

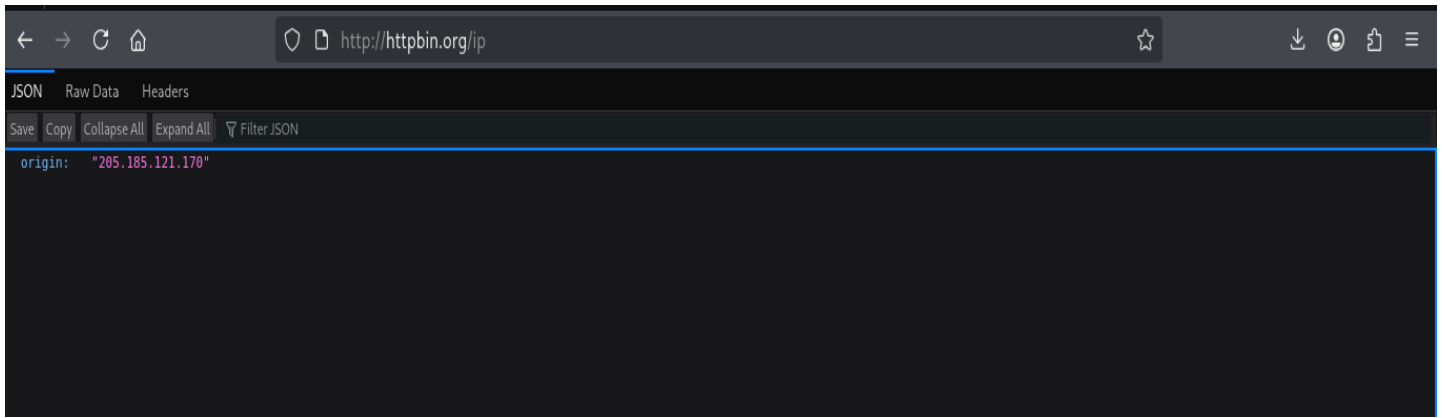
```

What do the results of this test mean?

```
sudo service tor@default restart
```

```
proxychains4 curl http://httpbin.org/ip
```

Result:



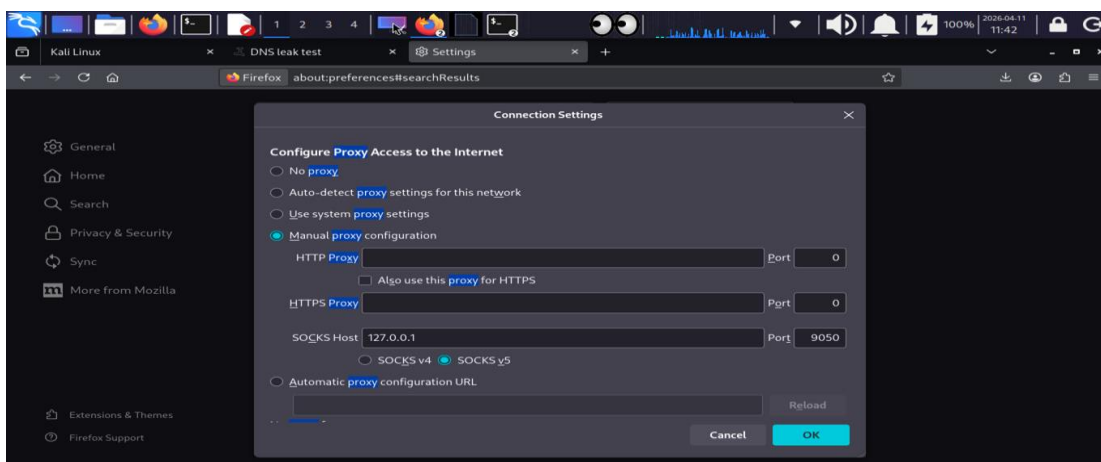
## 5. Firefox Proxy Configuration

To test browser-level anonymity:

### Steps:

- Open Firefox → Settings → Network Settings
- Select Manual Proxy
- Set:
  - SOCKS Host: 127.0.0.1
  - Port: 9050
  - SOCKS v5
  - Enable Proxy DNS

Result:



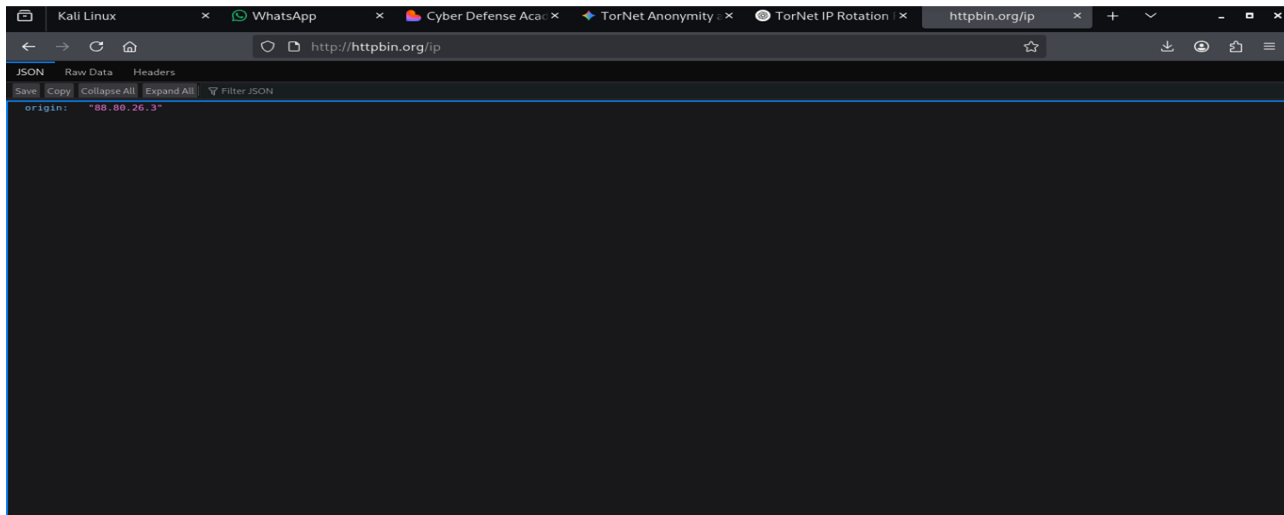
## 5.1 Browser IP Test

Visited:

- <http://httpbin.org/ip>

Confirmed Tor IP is visible in browser.

Result:



6.

## DNS Leak Testing

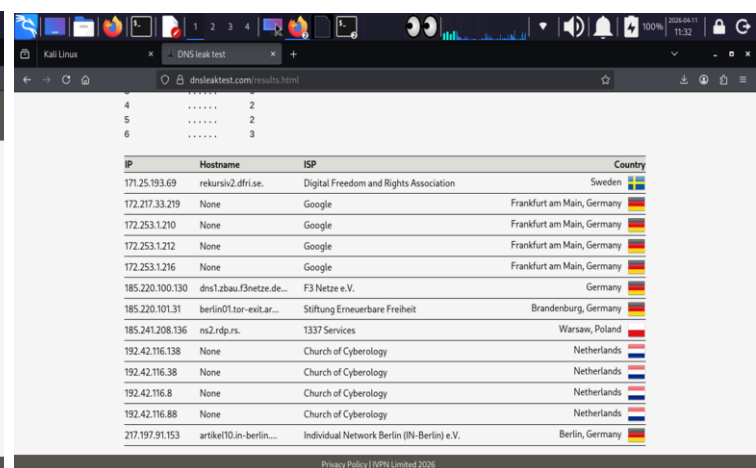
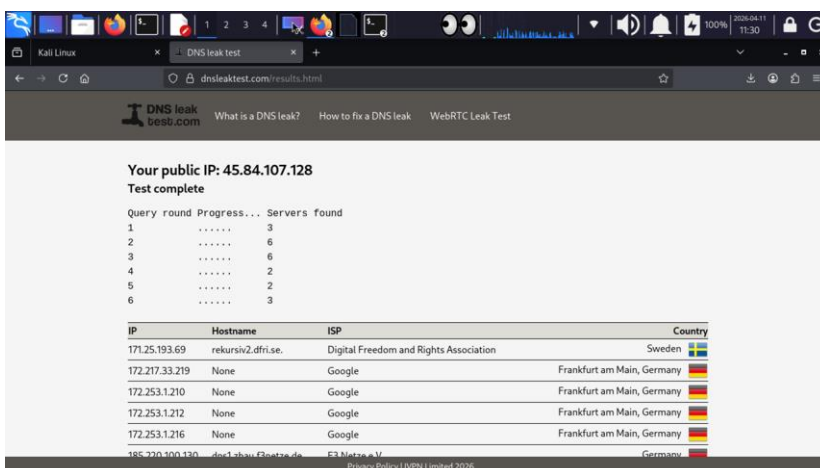
DNS leak testing was performed using:

DNSLeakTest

**Results:**

- Multiple DNS servers detected
- No ISP information visible
- Servers located in different regions

This confirms no DNS leak.



## 7. Automation Using Bash Script

To automate IP rotation:

```
#!/bin/bash

while true
do
 echo "Changing IP..."

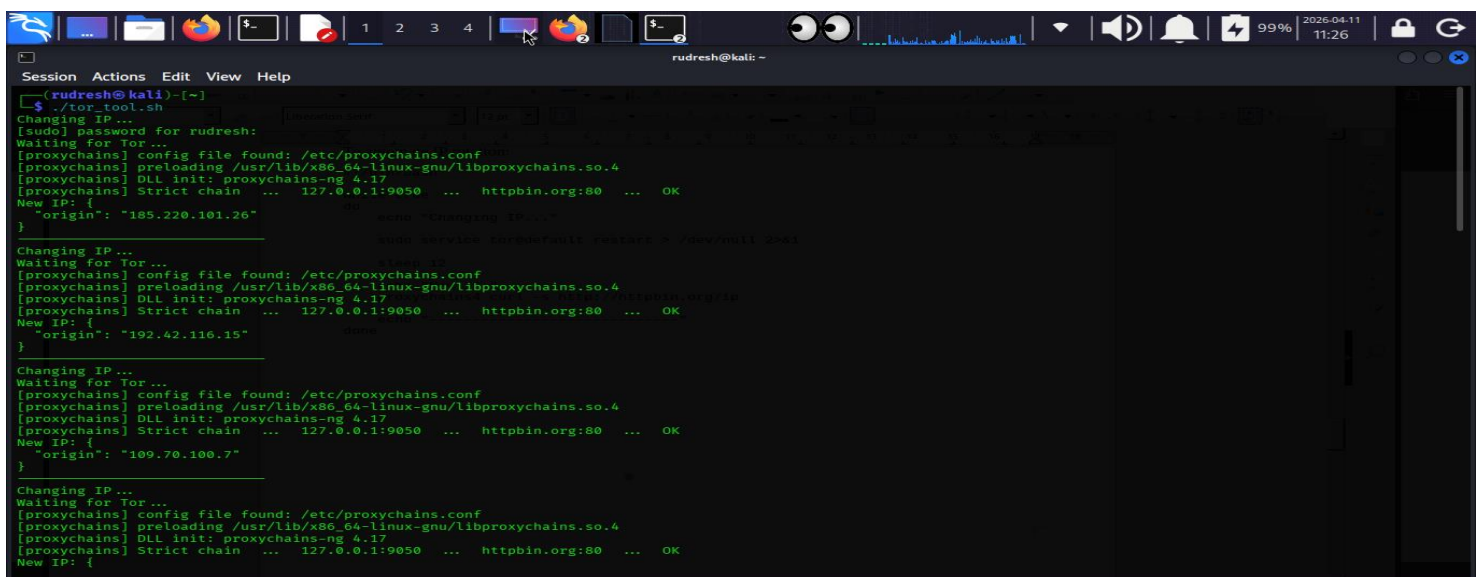
 sudo service tor@default restart > /dev/null 2>&1

 sleep 12

 echo "New IP:"
 proxychains4 curl -s http://httpbin.org/ip

 echo "-----"
done
```

Result:



```
Session Actions Edit View Help
(rudresh@kali)~$./tor_tool.sh
Changing IP...
[sudo] password for rudresh:
Waiting for Tor...
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:80 ... OK
New IP: {
 "origin": "185.220.101.26"
}
Changing IP...
[sudo] service tor@default restart > /dev/null 2>&1
Waiting for Tor...
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:80 ... OK
New IP: {
 "origin": "192.42.116.15"
}
Changing IP...
Waiting for Tor...
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:80 ... OK
New IP: {
 "origin": "109.70.100.7"
}
Changing IP...
Waiting for Tor...
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:9050 ... httpbin.org:80 ... OK
New IP: {
```

## 9. Operational Security (OPSEC)

### **Advantages:**

- Hides real IP
- Prevents tracking
- Useful in cybersecurity research

### **Limitations:**

- Slow network
- Exit node risks
- Not 100% anonymous

## 10. Skills Acquired

- Tor network configuration
- Proxychains usage
- Browser proxy setup
- DNS leak testing
- Bash scripting
- Debugging real issues

## 11. Conclusion

In this task, I successfully implemented anonymous communication using the Tor network. I verified IP anonymization and DNS protection. I also developed a script for automatic IP rotation, which improved operational security.

This task provided practical knowledge of anonymity techniques used in cybersecurity.

## 12. Final Outcome

- Tor configured successfully
- Anonymous browsing achieved
- IP rotation verified
- DNS leak prevented
- Automation implemented